A SUMMARY OF RESULTS AND PROBLEMS CONCERNING FLATNESS OF CODIMENSION ONE SPHERES IN E^n

BY

ROBERT J. DAVERMAN*

This article, which is devoted to summarizing the status of results and problems concerning the flatness of (n-1)-manifolds embedded in n-manifolds, represents an expanded version of a talk delivered at the Special Session on Geometrical Topology at the annual meeting of the American Mathematical Society in January, 1974. An unabashed imitation of Burgess's expository paper [10], this paper focuses in particular on the high dimensional analogues of the 3-dimensional work he discusses. Another extremely valuable and even more detailed description of the 3-dimensional situation is Burgess and Cannon's expository article [12].

No attempt has been made here to be exhaustive in listing references or results. The reader is invited to consult [10] and [12] for references to the related 3-dimensional work.

NOTATION. Throughout the paper S will denote an $(n-1)$-sphere topologically embedded in Euclidean n-space E^n $(n \geq 5)$, and Int S (Ext S) will denote the bounded (unbounded) component of $E^n - S$.

1. Results Implying Flatness.

1.1. If S is bicollared, it is flat [3].

1.2. If S is locally flat, it is flat [4].

1.3. If S is locally flat modulo one point, it is flat [15].

1.4. If S is locally flat except at the points of a set E, then E contains no isolated points [18], [35], [38].

1.5. If S is locally flat modulo a Cantor set X and X is tame both in S and in E^n, then S is flat [37].

1.6. If $E^n - S$ is 1-ULC, then S is flat [19], [22].

1.7. If S can be homeomorphically approximated from both components of $E^n - S$, then S is flat.

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1.8. If S is locally spanned from each component of $E^n - S$, then S is flat. (Burgess's proof in [11] shows that $E^n - S$ is 1-ULC.)

1.9. If S can be deformed into each component of $E^n - S$, then S is flat [27].

1.10. If there exists a curvilinear triangulation T of S for which each closed simplex of T is locally flatly embedded, then S is flat [17].

1.11. If S is locally flat modulo a k-cell K $(k \neq n - 3)$ such that K is flat both as a subset of S and as a subset of E^n, then S is flat (see [16] or show directly that $E^n - S$ is 1-ULC).

1.12. Suppose that S is locally flat modulo an (n-3)-cell K in S for which there exists an embedding f of $K \times [-1,1]$ onto a flat (n-2)-cell in E^n such that $S \cap f(K \times [-1,1]) = f(K \times 0) = K$ and $f(K \times [-1,0))$ and $f(K \times (0,1])$ lie in different components of $E^n - S$. Then S is flat [1].

1.13. Suppose that for each point x in S and each $\varepsilon > 0$ there exists an (n-1)-sphere S' in E^n such that $x \in \operatorname{Int} S'$, $\operatorname{diam} S' < \varepsilon$, and each component of $S' - (S' \cap S)$ is simply connected. Then S is flat [13].

1.14. Suppose that, for each (n-2)-cell K in S that is flat relative to S, S can be pierced along K by a flat (n-1)-cell (as in the hypothesis of 1.12). Then S is flat [28].

1.15. Suppose there exist an (n-1)-manifold M and a weak fibration $\phi : M \to S$ such that the mapping cylinder of ϕ can be naturally embedded in E^n so as to contain a neighorhood of S. Then S is flat [9].

1.16. Let $E_t = E^{n-1} \times \{t\}$ and $S_t = S \cap E_t$. Suppose S_1 and S_{-1} are points and S_t $(t \in (-1,1))$ is an (n-2)-sphere that is bicollared in both S and E_t. Then S is flat [20].

2. Examples of Wild (n-1)-Spheres in E^n.

The easiest method of producing wild spheres is to suspend a lower dimensional example. One takes an (n-2)-sphere S' in E^{n-1} such that $E^{n-1} - S'$ fails to be 1-ULC and defines S to be the natural suspension of $S' \times 0 \subset E^{n-1} \times 0 \subset E^n$ from the points $(0,\ldots,0, \pm 1)$. One can readily determine that $E^n - S$ fails to be 1-ULC and, hence, S cannot be flat.

A second method is to construct S so it contains a wild Cantor set. Given any Cantor set C in E^n, one can build a sphere S containing C so that S is locally flat modulo C (see [40]). If C is wild, it follows from the classical Klee trick [39] that S is wild.

As a third method, one can find wild spheres in the product of the line and certain decompositions G of E^{n-1} for which $E^{n-1}/G \times E^1 = E^n$. This method was exploited by Brown [5] and later by Rushing [42], where the decompositions G considered had a noncellular arc as the only nondegenerate element and the wild sphere (or cell) contained the image of the product of this nondegenerate element and an interval.

Besides the spheres constructed by the above techniques, there exist some examples having specific properties worth mentioning. There exist (see [26]) higher dimensional versions of Alexander's Horned Sphere, namely, wild (n-1)-spheres in E^n that are locally flat modulo a tame (relative to E^n) Cantor set. Such examples illustrate the sharpness of Kirby's theorem [37], mentioned here in 1.5. Also, there exists (see [24]) an (n-1)-sphere S in E^n such that every 2-cell in S is wildly embedded; hence, every k-dimensional ($k \geq 2$) polyhedron in S is wildly embedded in E^n. In addition, in contrast with the 3-dimensional situation described in [36], there exists a wild (n-1)-sphere S in E^n ($n \geq 4$) such that each straight line parallel to the x_n-axis intersects S in at most two points [29].

3. PROPERTIES OF SPHERES IN E^n.

According to Price-Seebeck [41], S can be **LOCALLY APPROXIMATED BY LOCALLY FLAT EMBEDDINGS** if for each point x of S there exists a neighborhood U of x in S such that the inclusion $U \to E^n$ can be approximated by locally flat embeddings of U. They prove the following two results:

3.1. If S can be locally approximated by locally flat embeddings, then S can be approximated by locally flat embeddings.

3.2. If Int S is 1-ULC and S can be (locally) approximated by locally flat embeddings, then S is collared from Int S.

The principle result of [41] is that S is flat if it is locally flat at some point and $E^n - S$ is 1-ULC, which was essential in one of the proofs of 1.6. In the other proof of 1.6, the following result was employed:

3.3. If there exists an (n-1)-dimensional Sierpiński curve X such that $S \subset X \subset \mathrm{Cl}(\mathrm{Int}\ S)$, then S is collared from Int S [14], [19].

3.4. Let A be an arc in S and $\varepsilon > 0$. There exists an embedding h of A in S that moves no point as much as ε such that $h(A)$ is flat relative to E^n [45]. On the basis of the following property, h can be obtained as the restriction of an ε-push of (S,A).

3.5. There exists a 1-dimensional F_σ set F in S such that $F \cup \mathrm{Int}\ S$ is 1-ULC [21].

3.6. There exists a 1-dimensional G_δ set G in S such that for each $\varepsilon > 0$ there is a map f of $\mathrm{Cl}(\mathrm{Int}\ S)$ into $G \cup \mathrm{Int}\ S$ such that f moves no point as much as ε and f is the identity outside the ε-neighborhood of S [21].

3.7. For $n \geq 6$ the following statements are equivalent:

(a) There exist curvilinear triangulations R of S of arbitrarily small mesh for which the 2-skeleta $R^{(2)}$ are tame relative to E^n.

(b) There exists a 0-dimensional F_σ set F in S that is a countable union of Cantor sets, each being tame relative to S, such that for each component U of $E^n - S$, $U \cup F$ is 1-ULC.

(c) For each k-dimensional $(k \leq n-3)$ polyhedron P topologically embedded in S and $\varepsilon > 0$, there is an ε-push h of (S,P) such that $h(P)$ is tame relative to E^n [7], [25].

3.8. If S is the suspension of an $(n-2)$-sphere in E^{n-1}, then each arc in S is flat and the statements of 3.7 all hold. Generally, if S is a k-fold suspension, each k-dimensional polyhedron in S is tame relative to E^n [23].

3.9. Suppose there exists a 0-dimensional F_σ set F in S such that for each component U of $E^n - S$, $U \cup F$ is 1-ULC. Then S can be approximated by locally flat embeddings. In fact, when there exists a 0-dimensional F_σ set F in S such that $F \cup \text{Ext } S$ is 1-ULC, then for each $\varepsilon > 0$ there exists an embedding h of $\text{Cl}(\text{Int } S)$ in E^n that moves no points as much as ε and $E^n - h(\text{Cl}(\text{Int } S))$ is 1-ULC [8].

4. Sewings of Crumpled Cubes.

A **crumpled n-cube** C is a space homeomorphic to the closure of a complementary domain of an $(n-1)$-sphere S topologically embedded in S^n. The set of points corresponding to S is referred to as the **boundary** of C (Bd C), and C - Bd C is called the **interior** of C (Int C).

By a **sewing** of two crumpled n-cubes C and D under a homeomorphism h of Bd C to Bd D, we mean the quotient space $C \underset{h}{\cup} D$ obtained from the disjoint union of C and D under the relation which identifies each point x in Bd C with $h(x) \in$ Bd D. We often refer to the homeomorphism h itself as a sewing of C to D. According to the generalized Poincaré Conjecture,

4.1. If $C \underset{h}{\cup} D$ is a manifold, it is topologically S^n.

Information about when a sewing of two crumpled n-cubes yields S^n falls far short of the precise characterization given for the case $n = 3$ by Eaton's Mismatch Theorem [32]. Indeed, it is not even known whether each crumpled n-cube can be sewn to an n-cell to produce S^n. Consequently, to simplify matters we shall adopt Cannon's terminology [14] and refer to a crumpled n-cube C as a **closed n-cell complement** if there exists an embedding h of C in S^n such that $\text{Cl}(S^n - h(C))$ is an n-cell.

4.2. There is a sewing of two crumpled n-cubes (closed n-cell-complements) that does not yield S^n. (Eaton's decomposition [33] that does not yield S^n can be construed as a sewing of a closed n-cell-complement to itself.)

4.3. Suppose that C_1 and C_2 are closed n-cell-complements such that there exist 0-dimensional F_σ sets F_i in Bd C_i, each of which is a countable union of Cantor sets that are tame relative to Bd C_i and $F \cup \text{Int } C_i$ is 1-ULC. Then there exists a sewing h of C_1 to C_2 such that $C_1 \underset{h}{\cup} C_2$ is homeomorphic to S^n. (The argument for the 3-dimensional case in [30] goes through under these stringent hypotheses.)

4.4. If h is a sewing of the closed n-cell-complements C and D, then the suspension of $C \underset{h}{\cup} D$ is homeomorphic to S^{n+1}. (In [31] it was observed that this fact can be established by a one directional squeeze based on Bryant's techniques [6].)

4.5. For any closed n-cell complement C bounded by one of the wild spheres of [26] that are locally flat modulo a tame Cantor set, the identity sewing of C to itself yields S^n [29].

A crumpled n-cube C is said to be UNIVERSAL if for each crumpled n-cube D and each sewing h of C to D, $C \underset{h}{\cup} D$ is homeomorphic to S^n. Similarly, C is said to be SELF-UNIVERSAL if each sewing of C to itself yields S^n. Virtually nothing is known about these properties for $n \geq 4$, and the purpose of this entire section has been to show, if only by omission, the lack of knowledge about sewings of crumpled n-cubes.

5. CONJECTURES.

5.1. For each component U of $E^n - S$ there exists a 0-dimensional F_σ set F in S such that $U \cup F$ is 1-ULC.

5.2. The n-cell B^n is a universal crumpled n-cube.

5.3. S can be homeomorphically approximated by locally flat spheres.

5.4. If Int S is 1-ULC, then S is collared from Int S.

REMARK. 5.1 ⇒ 5.2 ⇒ 5.3 ⇒ 5.4.

5.5. S can be almost approximated from either side, that is, for each component U of $E^n - S$ and $\varepsilon > 0$ there exists an ε-homeomorphism h of S in E^n and there exists a finite collection $D_1, \ldots, D_k$ of open, pairwise disjoint, ε-sets in S such that $h(S) \cap S$ is contained in $\cup D_i$ and the diameter of each component of $h(S) - U$ is less than ε. It should be mentioned that, unlike what occurs in Bing's 3-dimensional Side Approximation Theorem [2], one cannot require the D_i's to be open (n-1)-cells in S [25]. The combination of 5.3 and 5.5 suggests: S can be almost approximated from either side by locally flat spheres.

5.6. S is flat if it is free, that is, if for each component U of $E^n - S$ and each $\varepsilon > 0$, there is an ε-map of S into U.

5.7. Suppose for each point x in S, each component U of $E^n - S$, and each $\varepsilon > 0$, there exists an (n-1)-cell R in S such that $x \in$ Int R, diameter $R < \varepsilon$, and the identity map of Bd R is homotopic to a constant map in an ε-subset of $U \cup$ Bd R. Then S is flat.

5.8. Suppose for each point x in S and $\varepsilon > 0$ there exists an (n-1)-sphere S' in E^n such that $x \in$ Int S', diameter $S' < \varepsilon$, and $S \cap S'$ is connected. Then S is flat.

5.9. S is flat if it is homogeneous, that is, if for any two points x, y in S there is a homeomorphism H of E^n to itself such that $H(S) = S$ and $H(x) = y$.

5.10. S is flat if it is strongly homogeneous, that is, if each homeomorphism h of S to itself extends to a homeomorphism H of E^n to itself.

5.11. S is flat if it can be locally spanned in each complementary domain, that is, if for each point x in S, each component U of $E^n - S$, and each $\varepsilon > 0$, there exists an $(n-1)$-cell R in S such that $x \in \text{Int } R$, diameter $R < \varepsilon$, and for each $\alpha > 0$ there exists an $(n-1)$-cell D in U such that diameter $D < \varepsilon$ and Bd R is contractible in $D \cup$ (α-neighborhood of Bd R).

5.12. S is flat if it can be uniformly locally spanned in each complementary domain, that is, if the properties of 5.11 hold for all sufficiently small cells R in S.

5.13. S is flat if there exist an $(n-1)$-manifold M and a map $g : M \to S$ such that the mapping cylinder of g can be naturally embedded in E^n so as to contain a neighborhood of S.

5.14. S is flat if the intersection of every horizontal $(n-1)$-plane E_t with S is either a point or an $(n-2)$-sphere that is flat in E_t.

5.15. If X is a compact subset of S such that $E^n - S$ is 1-ULC in $E^n - X$, then X lies on some flat sphere in E^n.

5.16. For each $\varepsilon > 0$ there exists an $(n-2)$-dimensional continuum X in S such that each component of $S - X$ has diameter less than ε and X lies on some flat sphere in E^n.

5.17. (J. W. Cannon) Suppose X is a compact subset of S. There exists an embedding f of S in E^n such that $f(S)$ is locally flat modulo $f(X)$ and $f(S)$ is wild at each point $f(x) \in f(X)$ for which $S - X$ fails to be 1-LC at x.

5.18. There exists an $(n-1)$-sphere S in E^n such that each 2-cell in S is wildly embedded in E^n but each arc (Cantor set) in S is tame relative to E^n.

5.19. There exists a monotone map f of E^n to itself such that $f(E^n - S) = E^n - f(S)$, $f \mid S$ is one-one, and $f(S)$ is flat.

5.20. Suppose that, for each arc A in the boundary of a crumpled n-cube C, $C - A$ is 1-ULC. Then Bd C contains a 0-dimensional F_σ set F such that $F \cup \text{Int } C$ is 1-ULC.

5.21. For any two closed n-cell-complements C and D there exists a sewing of C to D that yields S^n.

5.22. Mismatch Conjecture: let h be a sewing of the closed n-cell-complements C_1 and C_2. Then $C_1 \underset{h}{\cup} C_2$ is S^n iff there exist sets F_i in Bd C_i such that $F_i \cup \text{Int } C_i$ is 1-ULC $(i = 1,2)$ and $h(F_1) \cap F_2 = \phi$.

5.23. A closed n-cell-complement C is a universal crumpled n-cube if, for each 1-dimensional compactum X in Bd C, $C - X$ is 1-ULC.

5.24. The suspension of a universal (self-universal) crumpled n-cube is universal (self-universal).

5.25. Like the Alexander Horned Crumpled Cube, the (non-cell) crumpled n-cubes of [26] are self-universal but not universal.

REMARK. At this time, 5.6, 5.7, 5.9, 5.10, and 5.11 all apparently remain unsolved for the case $n = 3$.

UNIVERSITY OF TENNESSEE
KNOXVILLE, TENNESSEE

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